Light

GCSE Computer Science – Computational Thinking (Set 2)

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GCSE • AQA 8525 • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (AQA 8525)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

Which of the following is an essential characteristic of an algorithm?

- ○ **A.** It must be written in a programming language
- ○ **B.** It must run on a computer
- ○ **C.** It must produce the same output every time for the same input
- ○ **D.** It must be kept secret from other programmers

Q2 (1 mark)

Which of the following is the best example of an algorithm in everyday life?

- ○ **A.** A photograph of a completed jigsaw puzzle

- ○ **B.** Step-by-step directions from a GPS navigation system to reach a destination
- ○ **C.** The ingredients list for a cake
- ○ **D.** A painting hanging in an art gallery

Q3 (1 mark)

Why is decomposition useful when developing a complex computer system?

- ○ **A.** It allows different parts to be developed and tested independently before being combined
- ○ **B.** It reduces the amount of storage needed
- ○ **C.** It makes the final program run faster
- ○ **D.** It automatically removes bugs from the code

Q4 (1 mark)

A weather forecasting system uses abstraction. Which approach best demonstrates this?

- ○ **A.** Including every cloud formation, individual raindrop, and blade of grass in the model
- ○ **B.** Making the forecast display more colourful
- ○ **C.** Focusing on key factors like temperature, pressure, humidity, and wind patterns while ignoring insignificant details
- ○ **D.** Running the forecast calculations more slowly for accuracy

Q5 (1 mark)

What is the main purpose of pseudocode?

- ○ **A.** To create the final working program
- ○ **B.** To encrypt algorithms so others cannot understand them
- ○ **C.** To make programs run faster than normal code
- ○ **D.** To express an algorithm in a structured, human-readable way without being tied to a specific programming language

Q6 (1 mark)

In a flowchart, what does a rectangle (box) shape represent?

- ○ **A.** The start or end of the algorithm
- ○ **B.** A decision where the flow branches
- ○ **C.** A process, calculation, or action being performed
- ○ **D.** User input being received

Q7 (1 mark)

In the algorithm statement 'OUTPUT "Your total is: " + total', what type of operation is being performed?

- ○ **A.** Input - data is being received from the user
- ○ **B.** Processing - a calculation is being performed
- ○ **C.** Storage - data is being saved to a file

- ○ **D.** Output - data is being displayed to the user

Q8 (1 mark)

What is a key advantage of linear search over binary search?

- ○ **A.** Linear search is always faster
- ○ **B.** Linear search works on unsorted data without requiring the list to be in order first
- ○ **C.** Linear search uses less memory
- ○ **D.** Linear search can only find numbers, not text

Q9 (1 mark)

What happens after each comparison in binary search?

- ○ **A.** The search moves to the next item in sequence
- ○ **B.** Approximately half of the remaining items are eliminated from consideration
- ○ **C.** All items are checked simultaneously
- ○ **D.** The list is re-sorted before continuing

Q10 (1 mark)

Why is bubble sort called 'bubble' sort?

- ○ **A.** Because larger values gradually 'bubble up' towards the end of the list through successive swaps
- ○ **B.** Because it was invented by someone named Bubble
- ○ **C.** Because it creates bubbles in memory
- ○ **D.** Because it makes a bubbling sound when running

Q11 (1 mark)

What additional resource does merge sort typically require compared to bubble sort?

- ○ **A.** Additional memory space to hold sublists during the merging process
- ○ **B.** A faster processor
- ○ **C.** An internet connection
- ○ **D.** A graphical display

Q12 (1 mark)

Two different algorithms both correctly sort a list of names into alphabetical order. What does this demonstrate?

- ○ **A.** One of the algorithms must be wrong
- ○ **B.** The algorithms must be identical internally
- ○ **C.** Alphabetical sorting is impossible

○ **D.** More than one algorithm can solve the same problem, though they may differ in efficiency

Q13 (2 marks)

Consider the algorithm:

```
count ← 0
FOR i ← 1 TO 3
   count ← count + i
ENDFOR
What is the final value of count?
```

○ **A.** 3

○ **B.** 9

○ **C.** 6

○ **D.** 0

Q14 (2 marks)

A programmer attempts to use binary search on an unsorted list containing [7, 2, 9, 4, 1, 8, 3]. What problem will occur?

○ **A.** The search will work perfectly but take longer

○ **B.** The computer will crash immediately

○ **C.** The list will automatically sort itself

○ **D.** The algorithm may incorrectly report that an item is not present when it actually is in the list, because eliminating halves based on the middle value is meaningless in unsorted data

Q15 (2 marks)

In one pass of bubble sort through a list of 6 items, how many comparisons of adjacent pairs are made?

- ○ **A.** 5 comparisons
- ○ **B.** 6 comparisons
- ○ **C.** 3 comparisons
- ○ **D.** 12 comparisons

Q16 (2 marks)

During the merge phase of merge sort, two sorted sublists [3, 7] and [2, 5] need to be combined. What is the correct merged result?

- ○ **A.** [3, 7, 2, 5] - the lists are simply joined together
- ○ **B.** [2, 5, 3, 7] - each list keeps its original order but they alternate
- ○ **C.** [7, 5, 3, 2] - the lists are joined in reverse
- ○ **D.** [2, 3, 5, 7] - elements are compared and placed in order

Q17 (2 marks)

When would linear search need to examine every single item in a list?

- ○ **A.** Only when searching for the first item
- ○ **B.** Only when the list contains duplicate values
- ○ **C.** When the target item is the last in the list, or when the target

item is not present at all

○ **D.** Linear search never examines every item

Q18 (2 marks)

A school wants to create a website for students to check homework, view timetables, and contact teachers. Using decomposition, how should this project be organised?

○ **A.** Build everything as one large page with all features mixed together

○ **B.** Separate into distinct modules: homework system (viewing/submitting assignments), timetable display (schedules and room allocation), and communication system (messaging and contact forms) - each developed independently

○ **C.** Only create the homework feature and ignore the rest

○ **D.** Copy another school's website exactly

Q19 (2 marks)

A chess computer program uses abstraction. Which approach demonstrates appropriate abstraction for analysing the game?

○ **A.** Modelling the exact material each chess piece is made from (wood, plastic, marble)

○ **B.** Calculating the exact weight of each piece to determine move priority

○ **C.** Only considering the colour of the chess board squares

○ **D.** Tracking each piece's position, movement rules, and relative value while ignoring irrelevant physical details

Q20 (2 marks)

Algorithm X can sort a list of 100 items in 0.1 seconds. Algorithm Y can sort the same list in 0.5 seconds. Which algorithm is more time-efficient for this task?

○ **A.** Algorithm Y because it takes more time, meaning it is more thorough

○ **B.** Algorithm X because it completes the same task in less time

○ **C.** They are equally efficient because both successfully sort the list

○ **D.** Efficiency cannot be determined from this information

Q21 (2 marks)

A flowchart has a diamond containing 'Is temperature > 30?' with 'Yes' leading to 'OUTPUT "Hot"' and 'No' leading to another diamond 'Is temperature < 10?' with 'Yes' leading to 'OUTPUT "Cold"' and 'No' leading to 'OUTPUT "Moderate"'. What outputs when temperature is 20?

○ **A.** Moderate

○ **B.** Cold

○ **C.** Hot

○ **D.** Both Hot and Cold

Q22 (2 marks)

Consider the algorithm:

```
a ← 10
b ← 3
IF a > b THEN
   a ← a - b
ENDIF
result ← a × b
What is the final value of result?
```

- ○ **A.** 30
- ○ **B.** 3
- ○ **C.** 7
- ○ **D.** 21

Q23 (2 marks)

A flowchart describes how to calculate the average of three numbers. A Python program is written that implements this flowchart. What is the relationship between the flowchart and the program?

- ○ **A.** The flowchart is the algorithm and the program is an implementation of that algorithm in a specific language
- ○ **B.** The flowchart is obsolete once the program is written
- ○ **C.** The program and flowchart are identical
- ○ **D.** The flowchart runs faster than the program

Q24 (3 marks)

A hospital maintains a list of 100,000 patient records sorted by patient ID number. Staff frequently need to find specific patients by their ID.

Which search algorithm should be used and why?

- ○ **A.** Linear search because it is simpler and hospital staff may not understand binary search
- ○ **B.** Binary search because the data is already sorted by ID, allowing each comparison to eliminate half the remaining records, finding any patient in approximately 17 comparisons instead of potentially 100,000
- ○ **C.** Neither - 100,000 records is too many to search
- ○ **D.** Random search - pick records at random until finding the right one

Q25 (3 marks)

The list [5, 1, 4, 2] is sorted using bubble sort. How many complete passes are needed to fully sort the list?

- ○ **A.** 3 passes
- ○ **B.** 2 passes
- ○ **C.** 1 pass
- ○ **D.** 4 passes

Q26 (3 marks)

Merge sort is applied to the list [9, 3, 6, 1]. What is the sequence of operations?

- ○ **A.** Divide into [9, 3] and [6, 1], then into [9], [3], [6], [1]; merge [9] and [3] into [3, 9], merge [6] and [1] into [1, 6]; finally merge [3, 9] and [1, 6] into [1, 3, 6, 9]
- ○ **B.** Compare adjacent pairs repeatedly until sorted

- ○ **C.** Find the smallest element and move it to the front, repeat
- ○ **D.** Randomly shuffle until sorted

Q27 (3 marks)

A team is developing a mobile app for ordering cinema tickets. How should they apply both decomposition and abstraction to this project?

- ○ **A.** Build everything at once without planning, then remove features that do not work
- ○ **B.** Decomposition: separate into modules (film browsing, seat selection, payment processing, ticket delivery); Abstraction: model films by title/time/rating while ignoring irrelevant details like film reel material; model seats by row/number/availability while ignoring seat fabric colour
- ○ **C.** Only use abstraction and ignore decomposition entirely
- ○ **D.** Copy an existing cinema app without any original design

Q28 (3 marks)

A sorted list contains [10, 20, 30, 40, 50, 60, 70]. Binary search is used to find value 20. Which elements are compared during the search?

- ○ **A.** 10, then 20 - searching from the start
- ○ **B.** 70, 60, 50, 40, 30, 20 - searching from the end
- ○ **C.** 40 (middle), then 20 (middle of left half) - found in 2 comparisons
- ○ **D.** All seven elements are compared

Q29 (3 marks)

A small bakery has a list of 15 daily orders that need sorting by delivery time. The list is almost sorted - only 2 orders are in the wrong position. Which sorting algorithm would be most efficient and why?

- ○ **A.** Merge sort because it is always the fastest algorithm
- ○ **B.** Linear search because searching is faster than sorting
- ○ **C.** No sorting is needed for only 15 items
- ○ **D.** Bubble sort because it can detect the nearly-sorted state quickly, completing in very few passes when only minor corrections are needed

Q30 (3 marks)

A student needs to create a program that analyses text messages to count word frequency and identify the most common words. They have never built a text analysis tool before. What systematic approach should they use?

- ○ **A.** Start coding immediately without any planning
- ○ **B.** Search online for a complete solution and copy it without understanding
- ○ **C.** Use computational thinking: decompose into sub-problems (reading input text, splitting into words, counting occurrences, identifying most frequent); abstract by focusing on word strings and counts while ignoring message metadata like timestamps; design algorithms for each sub-problem before coding
- ○ **D.** Declare the problem impossible without trying

Total: **55** marks • Time: **30 min**

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